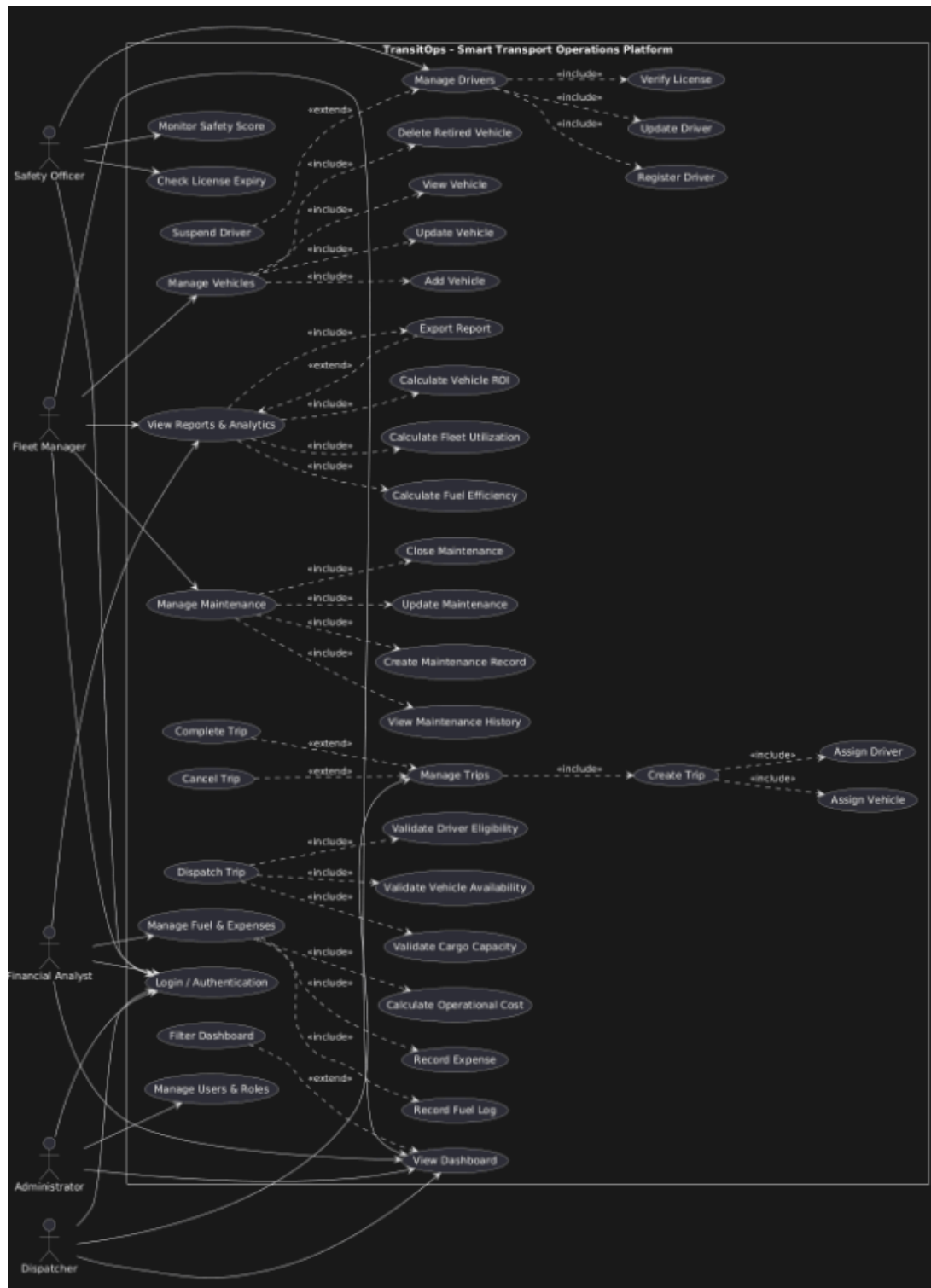


Aim

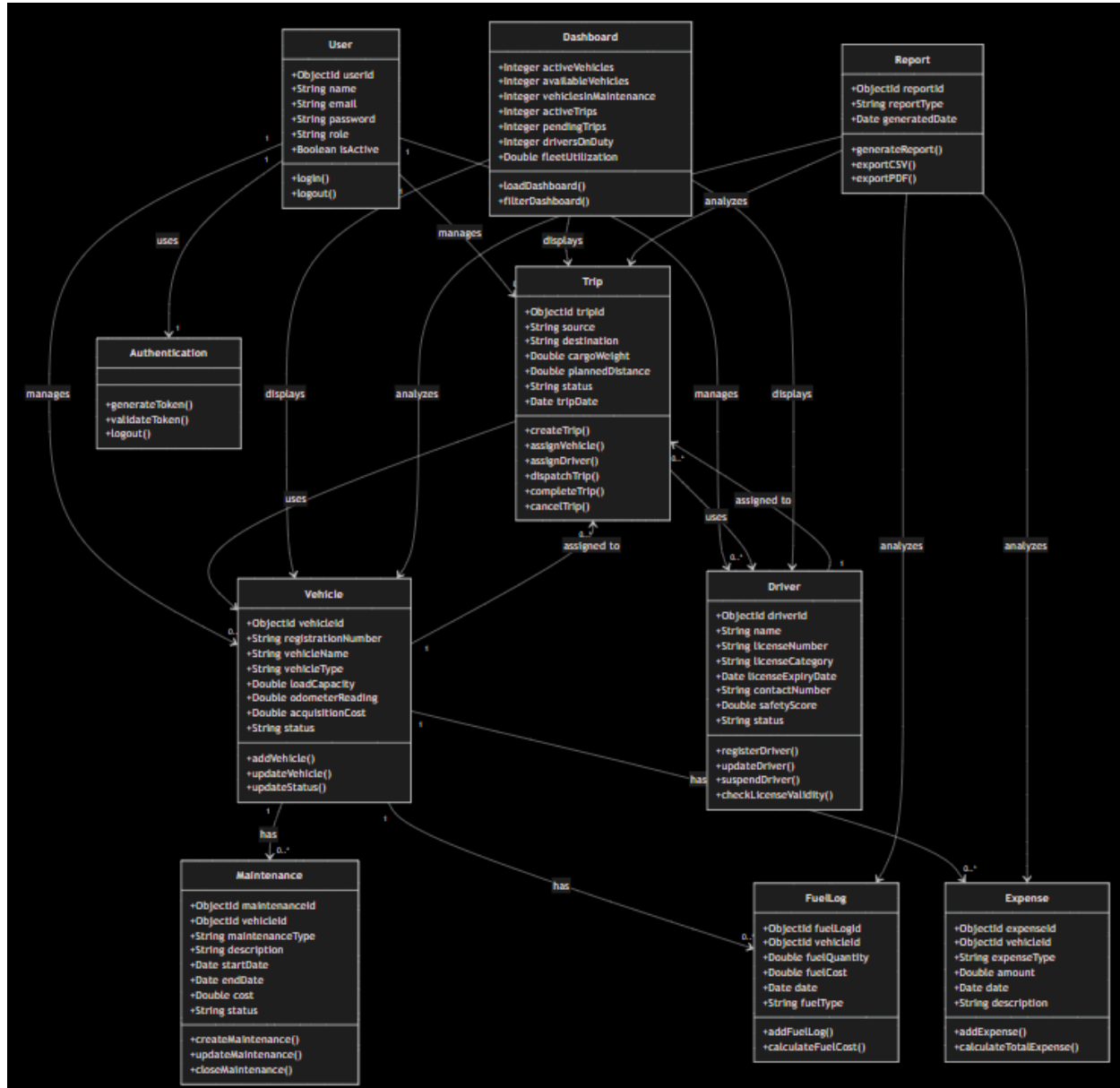
To design the architecture for the proposed application.

UML Diagrams

- Use Case Diagram

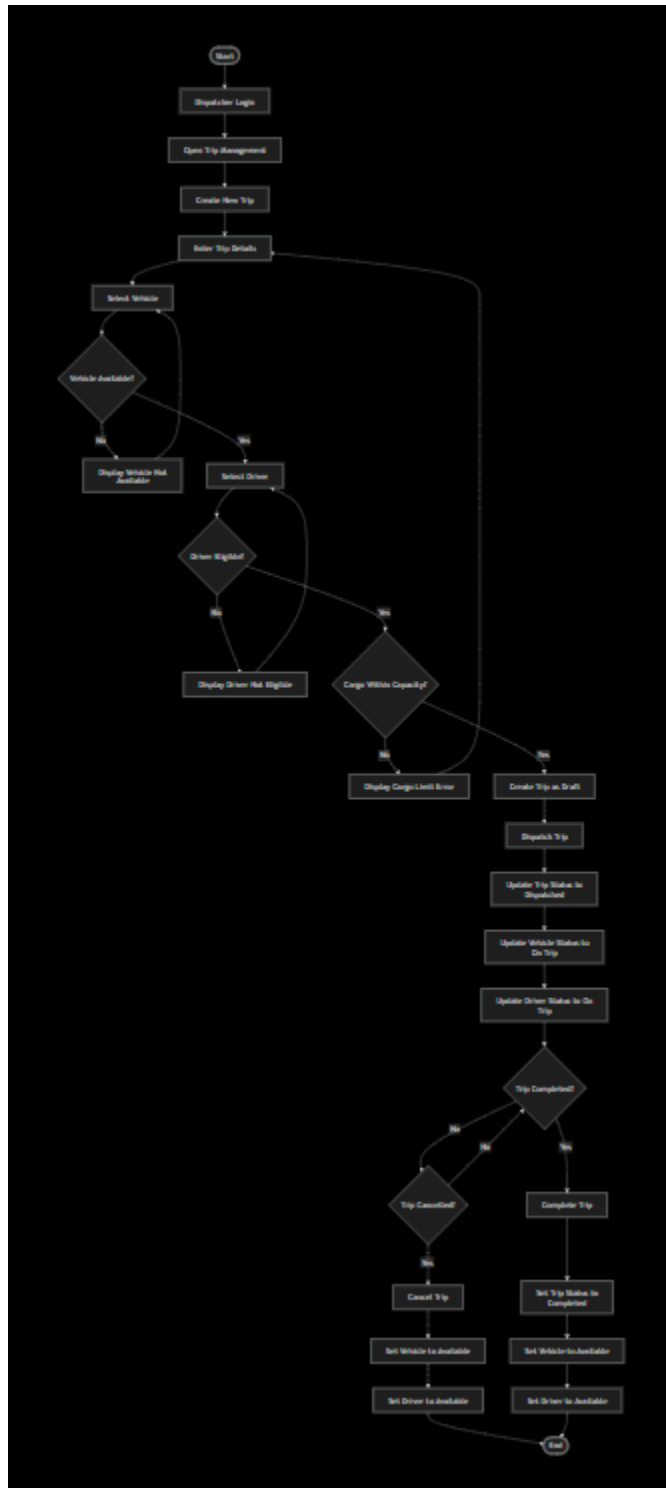


- Class Diagram

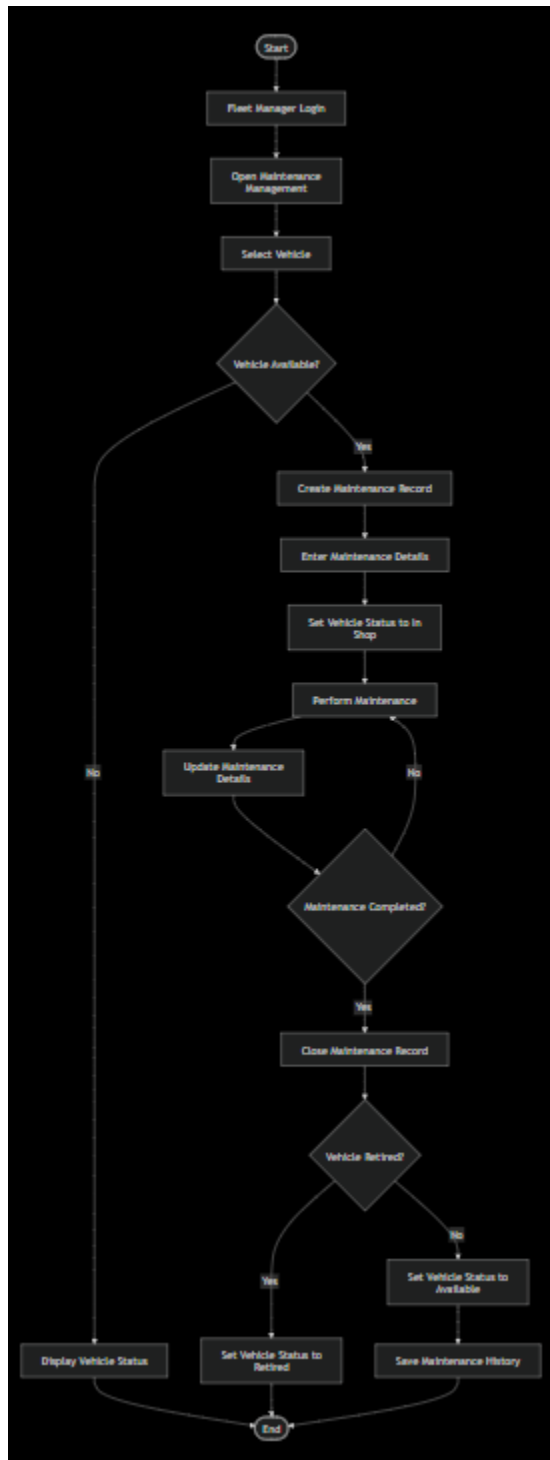


- Activity Diagrams

(i) Activity Diagram – Trip Management

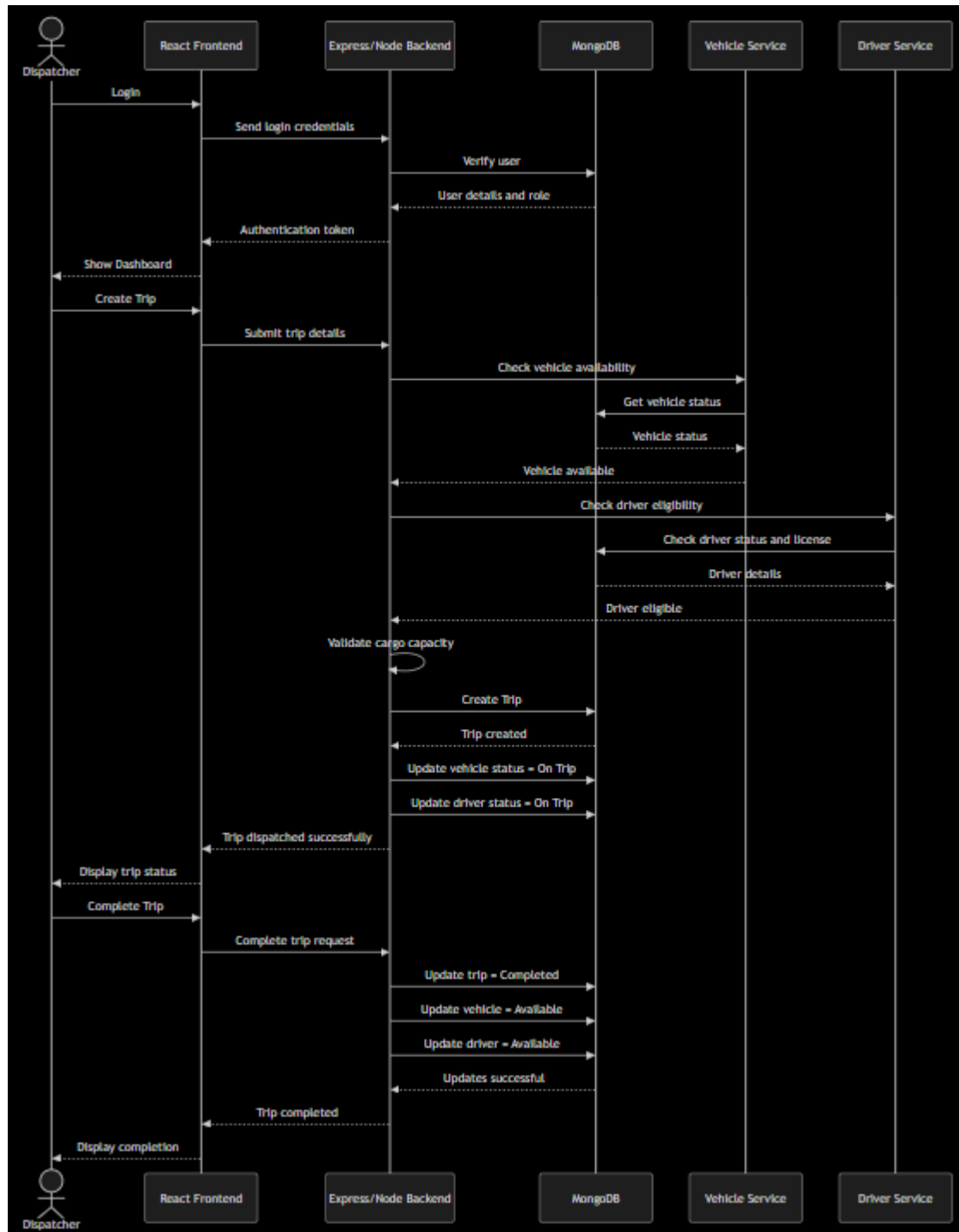


(ii) Activity Diagram – Maintenance Management



- Sequence Diagrams

(i) Sequence Diagram – Trip Management



(ii) Sequence Diagram – Maintenance Management



API Design

Prepare REST API documentation table containing the GET, POST, PUT and DELETE methods with URL and Description for each data entity of the system.

1. User API

Method	URL	Description
GET	/api/users	Retrieve all users
GET	/api/users/:id	Retrieve a specific user
POST	/api/users	Create a new user
PUT	/api/users/:id	Update user information
DELETE	/api/users/:id	Delete a user

Method	URL	Description
POST	/api/auth/login	Authenticate user and generate JWT token
POST	/api/auth/logout	Log out the authenticated user

2. Vehicle API

Method	URL	Description
GET	<code>/api/vehicles</code>	Retrieve all vehicles
GET	<code>/api/vehicles/:id</code>	Retrieve details of a specific vehicle
POST	<code>/api/vehicles</code>	Register a new vehicle
PUT	<code>/api/vehicles/:id</code>	Update vehicle information
DELETE	<code>/api/vehicles/:id</code>	Delete a retired vehicle

3. Driver API

Method	URL	Description
GET	<code>/api/drivers</code>	Retrieve all drivers
GET	<code>/api/drivers/:id</code>	Retrieve details of a specific driver
POST	<code>/api/drivers</code>	Register a new driver
PUT	<code>/api/drivers/:id</code>	Update driver information

DELETE	<code>/api/drivers/:id</code>	Delete a driver record
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4. Trip API

Method	URL	Description
GET	<code>/api/trips</code>	Retrieve all trips
GET	<code>/api/trips/:id</code>	Retrieve details of a specific trip
POST	<code>/api/trips</code>	Create a new trip
PUT	<code>/api/trips/:id</code>	Update trip information or status
DELETE	<code>/api/trips/:id</code>	Delete or remove a trip

Trip Operations

Method	URL	Description
PUT	<code>/api/trips/:id/dispatch</code>	Dispatch a trip
PUT	<code>/api/trips/:id/complete</code>	Mark a trip as completed
PUT	<code>/api/trips/:id/cancel</code>	Cancel a trip

5. Maintenance API

Method	URL	Description
GET	<code>/api/maintenance</code>	Retrieve all maintenance records
GET	<code>/api/maintenance/:id</code>	Retrieve a specific maintenance record
POST	<code>/api/maintenance</code>	Create a new maintenance record
PUT	<code>/api/maintenance/:id</code>	Update maintenance information
DELETE	<code>/api/maintenance/:id</code>	Delete a maintenance record

Maintenance Operations

Method	URL	Description
PUT	<code>/api/maintenance/:id/close</code>	Close a completed maintenance record
GET	<code>/api/vehicles/:id/maintenance</code>	Retrieve maintenance history of a vehicle

6. Fuel Log API

Method	URL	Description
GET	<code>/api/fuel-logs</code>	Retrieve all fuel records
GET	<code>/api/fuel-logs/:id</code>	Retrieve a specific fuel record
POST	<code>/api/fuel-logs</code>	Add a new fuel record
PUT	<code>/api/fuel-logs/:id</code>	Update a fuel record
DELETE	<code>/api/fuel-logs/:id</code>	Delete a fuel record

7. Expense API

Method	URL	Description
GET	<code>/api/expenses</code>	Retrieve all expense records
GET	<code>/api/expenses/:id</code>	Retrieve a specific expense
POST	<code>/api/expenses</code>	Record a new expense
PUT	<code>/api/expenses/:id</code>	Update an expense record

DELETE	<code>/api/expenses/:id</code>	Delete an expense record
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8. Dashboard API

Method	URL	Description
GET	<code>/api/dashboard</code>	Retrieve dashboard statistics
GET	<code>/api/dashboard/vehicles</code>	Retrieve vehicle-related dashboard statistics
GET	<code>/api/dashboard/trips</code>	Retrieve trip-related dashboard statistics
GET	<code>/api/dashboard/drivers</code>	Retrieve driver-related dashboard statistics
GET	<code>/api/dashboard/recent-activities</code>	Retrieve recent system activities

9. Reports and Analytics API

Method	URL	Description
GET	<code>/api/reports</code>	Retrieve available reports
GET	<code>/api/reports/fuel-efficiency</code>	Generate fuel efficiency report

GET	<code>/api/reports/fleet-utilization</code>	Generate fleet utilization report
GET	<code>/api/reports/vehicle-roi</code>	Calculate vehicle ROI
GET	<code>/api/reports/vehicle-performance</code>	Generate vehicle performance report
GET	<code>/api/reports/export/csv</code>	Export report in CSV format